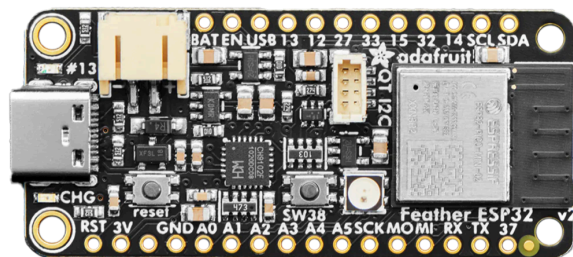
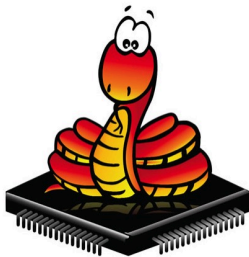


Artificial Intelligence of Things - Intelligent and Connected Systems

EECS E4764
Columbia University

Prof. Xiaofan Jiang

Setting Up Environment

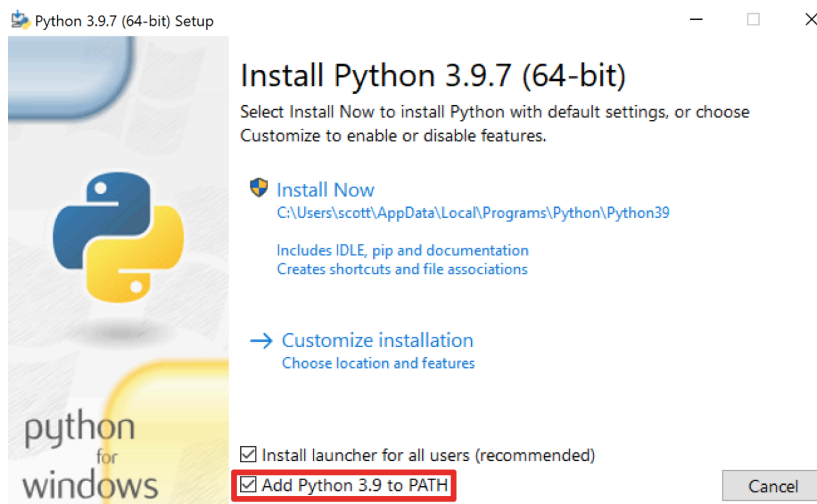


Windows

Part 1: Install Python

Step 1: Install

If you haven't installed Python on your computer, or installed Python from the Windows Store before, download and install Python 3 (3.13-3.14) from <https://www.python.org/downloads/>. Make sure "Add Python to PATH" is checked. Skip to step 2 if you already have Python installed.



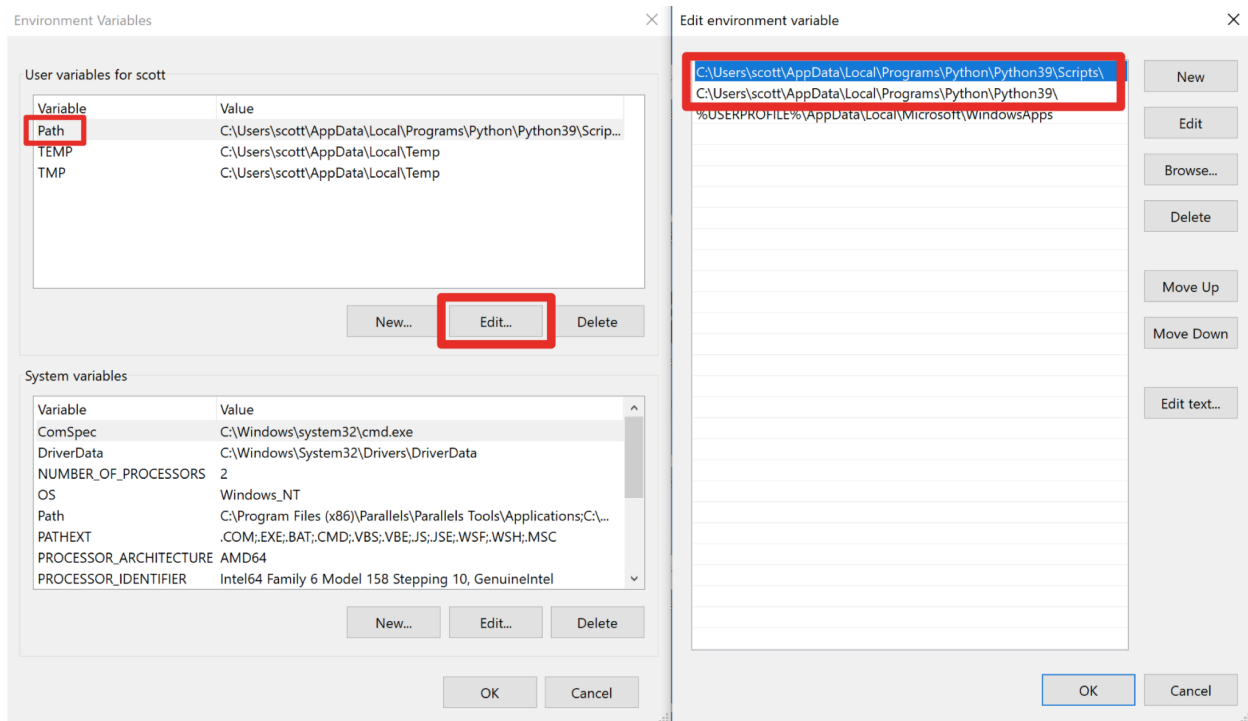
Step 2: Check PATH

Check your Python installation by opening a command prompt, and type

```
> python --version
> pip --version
```

If Windows Store opens up, or you encounter the error: 'python' / 'pip' is not recognized as an internal or external command ..., please continue below. If no error occurs for both commands, go to **Part 2: Flash HUZZAH Firmware**.

Make sure your environment variables are properly configured. Navigate to **Control Panel > System and Security > System > Advanced System Settings > Environment Variables**. Check to see if the path to your Python installation folder (where python.exe is located) and the path to the Python scripts folder (where pip.exe or pip3.exe is located) are in the "PATH" variable either in the user variables or the system variables. If they are not present, add the paths to the "PATH" user variable; create the "PATH" user variable if it does not already exist.



Close and reopen a command prompt, then verify again with

```
> python --version
```

```
> pip --version
```

Windows

Part 2: Flash HUZZAH Firmware

Step 1: Install Packages

Download the Micropython firmware from (**DO NOT UNZIP**)

https://micropython.org/resources/firmware/ESP32_GENERIC-20250809-v1.26.0.bin

Then, open a command prompt and type

```
> pip install esptool  
> pip install mpfshell
```

If you encounter the error: 'pip' is not recognized as an internal or external command,

...

Please check if your path is set up correctly. (Step 2 of last part) If you installed Python from Windows Store, please re-install it from the official website following Step 1 of the last part.

We will use esptool to flash MicroPython firmware downloaded above, and mpfshell to upload programs and communicate with HUZZAH.

Check the installation by typing in command prompt

```
> esptool  
> mpfshell
```

If you run into any issues with the installation, you should first find the installation location using Windows Explorer, and whether that folder is in the "PATH" variable (Part 1 Step 2). You can add that folder to "PATH", or replace the command with its absolute path.

i.e. `python <full\path\to>mpfshell.py` if it's a .py, or `<full\path\to>mpfshell` if it's a .exe.

You can also refer to their documentation on alternative ways to install:

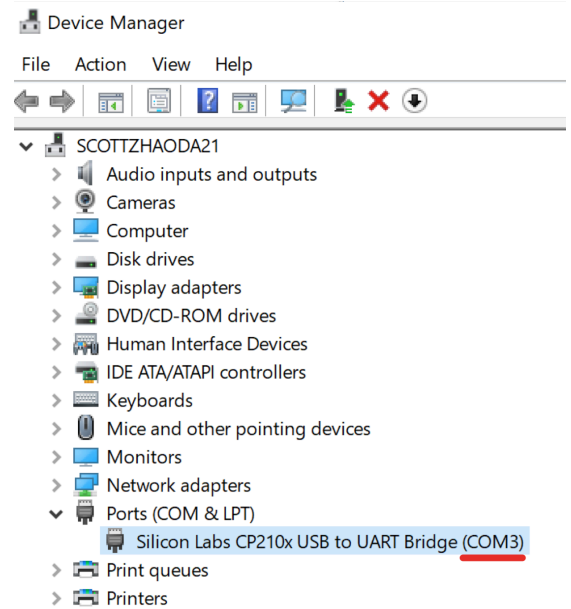
<https://github.com/espressif/esptool#installation--dependencies>

<https://github.com/wendlers/mpfshell#installing>

Step 2: Flash Firmware

Connect the HUZAH board to your machine and find out which serial port it is connected to. You can do this by looking under the Ports tab in **This PC → Properties → Device Manager → Ports**.

The device will be shown as USB-Enhanced-SERIAL CH9102 with a (COM*). If you see this, record the number after COM, and skip to the Flash MicroPython part below. **If you can't find this**, that means your computer is missing the driver.

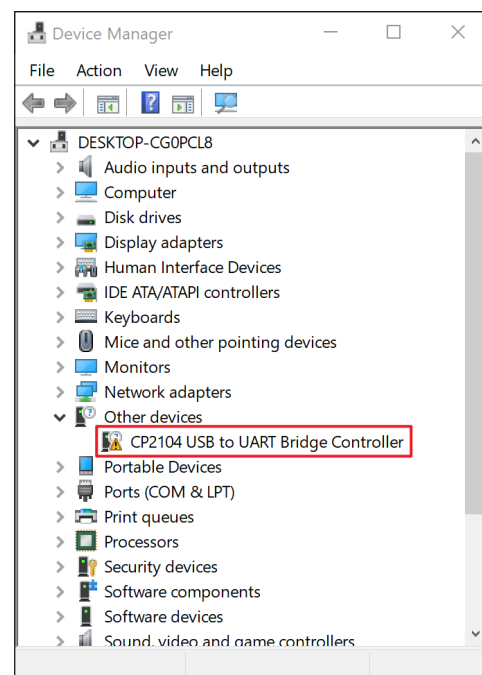


1)

To install the driver, first download the USB Driver (**for Universal Windows**) from https://www.silabs.com/documents/public/software/CP210x_Universal_Windows_Driver.zip

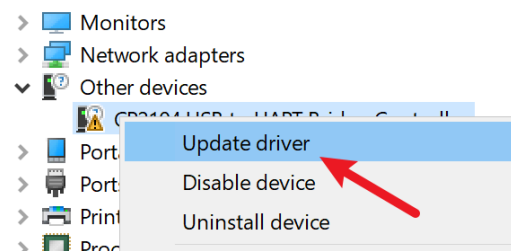
Then, unzip the zip file into a folder "CP210x_Universal_Windows_Driver".

Locate the device in *Device Manager*. It should be under "Other devices".



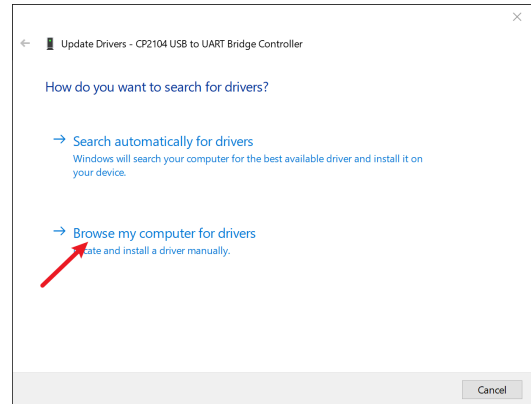
2)

Right click and choose "Update driver"



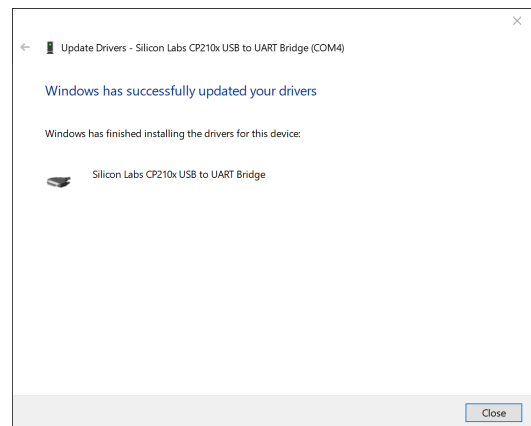
3)

Click “Browse my computer for drivers”



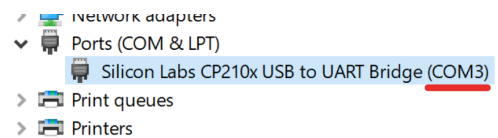
4)

Select the folder where you’ve unzipped the driver file, and click continue. Once the installation finishes, you should see this screen.



5)

Now you should see the device with a “COM*” number. Record this number and continue to the next step.



Flash MicroPython

Once you know the serial port, navigate to the folder through the command line to where the firmware was downloaded to. Enter the following command to flash the firmware onto the board. Each command may take up to a minute so please be patient.

```
> esptool --port COM* erase_flash
> esptool --port COM* --baud 460800 write_flash 0x1000 ESP32_GENERIC-20250809-v1.26.0.bin
```

After both commands are completed, click the “RST” (reset) button on your Feather HUZZAH.

Now that the firmware is set up on the HUZZAH, you can now connect to it through serial using a terminal program (mobaXterm, putty, termite, etc.) with the following settings:

```
Baudrate = 115200
Data bits = 8
Stop bits = 1
Parity = None
Flow Control = XON/XOFF
```

You should now be able to access the Python command line on the ESP8266 chip. The Micropython library has some, not all, of the same functionalities as Python 3, as well as some board-specific capabilities. More information can be found by looking through the documentation. We will show you how to program the microcontroller in the Lab document.

Step 3: Verify Firmware

Test connecting to your Feather HUZZAH using *mpfshell* we installed in the first step.

Open a command prompt and type

```
> mpfshell --open COM? -nc repl
```

You should be able to access the Python command line on the HUZZAH. You can try typing the following command to toggle a built-in LED on the HUZZAH.

```
MicroPython v1.20.0 on 2023-04-26; ESP module with ESP8266
Type "help()" for more information.
>>> from machine import Pin
>>> Pin(13, Pin.OUT).value(0)
>>> Pin(13, Pin.OUT).value(1)
```

This concludes the environment set-up on Windows. We will learn more about MicroPython in the lab.

Windows

Part 3: IDE (Optional)

If you are new to Python, or new to using Command Prompt to run Python code, you can use any existing MicroPython IDE to make things easier for you.

For reference, you can check out [Arduino Micropython IDE](#).

macOS / Linux

Part 1: Install Python

Step 1: Install

If you have previously used Python 3 (3.13-3.14) or it has already been installed, you can skip this step.

Otherwise, install Python 3 from
<https://www.anaconda.com/download>

When the installation is finished, restart your terminal and run the following command

```
> conda init
```

You should see the terminal changes to *(base) userName@laptopName ~ %* this means the default Python environment is successfully activated.

macOS / Linux

Part 2: Flash HUZZAH Firmware

Step 1: Install Packages

First, download the Micropython firmware from (**DO NOT UNZIP**)

https://micropython.org/resources/firmware/ESP32_GENERIC-20250809-v1.26.0.bin

Then, open a command prompt and type

```
> pip install esptool  
> pip install mpfshell
```

We will use esptool to flash MicroPython firmware downloaded above, and mpfshell to upload programs and communicate with HUZZAH.

Check the installation by typing in terminal

```
> esptool  
> mpfshell
```

If you run into any issues with the installation, you can refer to their documentation on alternative ways to install:

<https://github.com/espressif/esptool#installation--dependencies>

<https://github.com/wendlers/mpfshell#installing>

Step 2: Flash Firmware

Download and install the USB Driver from

<https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers>

(your device may already have a variation of this installed, and you don't need this specific version as long as you are able to see the device from the terminal command below)

Connect the HUZZAH board to your machine to check that it is recognized. Type in terminal

```
(macOS) > ls /dev/tty.*  
(Linux) > ls /dev/ttyUSB*
```

You should see /dev/tty.SLAB_USBtoUART for macOS and /dev/ttyUSB0 for Linux.

(Linux) If you see multiple ttyUSB*, keep going as esptool will figure out which one is our ESP32.

Now let's flash the firmware to our HUZZAH.

Using terminal, change directory into where you've downloaded the firmware file, and execute the following commands. Each command may take up to a minute so please be patient. (Linux) if you encountered permission error, follow this document to resolve it.

<https://websistent.com/fix-serial-port-permission-denied-errors-linux/>

```
> esptool erase_flash
```

```
> esptool --baud 460800 write-flash 0x1000 ESP32_GENERIC-20250809-v1.26.0.bin
```

After both commands succeed like below, click the "RST" (reset) button on your Feather HUZZAH. Take a note of this Serial port as we will use it in the next step.

```
(IoT_3.7) parallels@parallels-Parallels-Virtual-Platform:~$ esptool.py --baud 460800
write_flash --flash_size=32m 0 esp8266-20210902-v1.17.bin
WARNING: Flash size arguments in megabits like '32m' are deprecated.
Please use the equivalent size '4MB'.
Megabit arguments may be removed in a future release.
esptool.py v3.1
Found 1 serial ports
Serial port /dev/ttyUSB0
Connecting...
Detecting chip type... ESP8266
Chip is ESP8266EX
Features: WiFi
Crystal is 26MHz
MAC: 84:cc:a8:83:0d:7d
Uploading stub...
Running stub...
Stub running...
Changing baud rate to 460800
Changed.
Configuring flash size...
Flash will be erased from 0x00000000 to 0x0009afff...
Flash params set to 0x0040
Compressed 633688 bytes to 416262...
Wrote 633688 bytes (416262 compressed) at 0x00000000 in 9.9 seconds (effective 514.3
kbit/s)...
Hash of data verified.
Leaving...
```

Now that the firmware is set up on the HUZZAH, you can now connect to it through serial using a terminal program (mobaXterm, putty, termite, etc.) with the following settings:

Baudrate = 115200

Data bits = 8

Stop bits = 1

Parity = None

Flow Control = XON/XOFF

You should now be able to access the python command line on the ESP32 chip. The Micropython library has some, not all, of the same functionalities as Python 3, as well

as some board-specific capabilities. More information can be found by looking through the documentation. We will show you how to program the microcontroller in the Lab document.

Step 3: Verify Firmware

Test connecting to your Feather HUZZAH using *mpfshell* we installed in the first step.

Open a command prompt and type the following command. Replace the serial port with yours found in the previous step.

Note you have to remove `/dev/` from the path `/dev/tty****`

```
> mpfshell --open tty**** -nc repl
```

You should be able to access the Python command line on the HUZZAH. You can try typing the following command to toggle a built-in LED on the HUZZAH.

```
MicroPython v1.26.0 on 2025-08-09; Generic ESP32 module with ESP32  
Type "help()" for more information.
```

```
>>> from machine import Pin  
>>> Pin(13, Pin.OUT).value(0)  
>>> Pin(13, Pin.OUT).value(1)
```

This concludes the environment set-up on macOS / Linux. We will learn more about MicroPython in the lab.